

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (Credit System) Degree Examinations

January 2023

19ME8X28 – OPERATIONS MANAGEMENT AND ENTREPRENEURSHIP

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Explain how the production system is classified. | 10 | L*1 | 1 | 1 |
| b) Explain different cost of quality and its influence on the production management. | 10 | L1 | 2 | 1 |
| 2. a) Discuss the framework of operations management systems. | 10 | L1 | 1 | 1 |
| b) List seven basic tools used for quality control and explain any four. | 10 | L1 | 2 | 1 |
| 3. a) Write a note on: i) ERP ii) Kaizen | 10 | L2 | 1 | 1 |
| b) A machine shop produces steel pins. The width of 100 pins were tested after machining and the data was recorded as follows.
i) Find the arithmetic mean, standard deviation and variance.
ii) What percentage of the pins manufactured has width of 9.52 to 9.63 | | | | |

Width in mm	Frequency	Width in mm	Frequency
9.50-9.51	6	9.58-9.59	22
9.52-9.53	2	9.60-9.61	8
9.54-9.55	20	9.62-9.63	6
9.56-9.57	32	9.64-9.65	4

10 L1 2 2

Unit – II

- | | | | | |
|--|---|----|---|---|
| 4. a) Discuss the analysis involved in the Appraisal of a proposed project. | 8 | L1 | 4 | 1 |
| b) The following table shows the averages and ranges of the spindle of diameter 45.45 ± 0.5 in millimeters for 20 subgroups of 5 items each. | | | | |

$\bar{X}$	R	$\bar{X}$	R
45.020	0.375	45.600	0.275
44.950	0.450	45.020	0.175
45.480	0.450	45.320	0.200
45.320	0.150	45.560	0.425
45.280	0.200	45.140	0.250
45.820	0.250	45.620	0.375
45.580	0.275	45.800	0.475
45.400	0.475	45.500	0.200
45.660	0.475	45.780	0.275
45.680	0.275	45.640	0.225

Construct $\bar{X}$ and R chart and find out the process capability for the machine.

12 L2 3 2

P.T.O.

5. a) Write a note on Bathtub Curve.

10 L2 3 1

- b) A company has 3 factories manufacturing the same product and 5 sales agencies in different parts of the country. Production costs differ from factory to factory. The shipping cost per unit product from each factory to each agency is known. Find the production and distribution schedules which results in minimum cost.

Factory i	Production cost / unit (Rs.)	Max. capacity No. of units
1	18	140
2	20	190
3	16	115

Factory i	Agency j				
	1	2	3	4	5
1	2	2	6	1	5
2	10	8	9	4	7
3	5	6	4	3	8
Demand	84	104	79	39	139

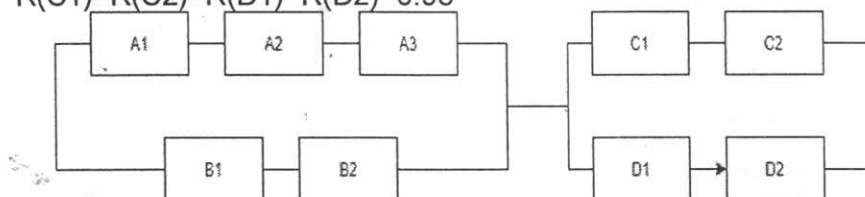
10 L2 4 2

6. a) What is the role that entrepreneurship play in the economic development of a country?

10 L1 4 1

- b) Consider the braking system of an automobile as shown in the figure. There are two partly independent systems. A pedal operated hydraulic system (A) and a mechanical system (B). Both actuate on the same set of brake shoes. For safe function, either both from wheels (C) brakes or both rear wheel wheels (D) brakes must function, otherwise automobile is uncontrollable. Calculate the Reliability of the overall system, if the individual reliabilities are given as follows.

$R(A1)=0.995$; $R(A2)=0.975$; $R(A3)=0.972$; $R(B1)=0.99$; $R(B2)=0.98$;
 $R(C1)=R(C2)=R(D1)=R(D2)=0.98$



10 L2 3 2

Unit – III

7. a) Discuss the Impact of Liberalization, Privatization and Globalization on SSI.

10 L1 5 1

- b) State the objectives and scope of small scale industry.

10 L1 5 1

8. a) Discuss the effect of WTO/GATT on Indian SSI.

10 L1 5 1

- b) List the supporting agencies of government for SSI.

10 L1 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
